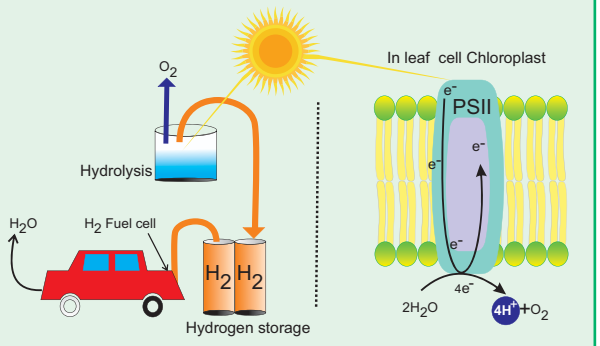
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A quest for future energy

Hydrogen is considered as a promising energy vector for the next generation. It can be used for "green" electricity production or developing cogeneration systems such as fuel cells. The sustainability of its employment depends on the energy source used to synthesize it from hydrogen-rich compounds such as water or biomass. The splitting of water in hydrogen and oxygen by means of solar radiation in Photolysis is common in plants. Water splitting is not an easy process to mimic artificially but preliminary success is achieved so far. If young minds take up this as their research ambition a revolution can be made in green energy.



The plants get energy from sun by converting solar or radiant energy into chemical energy by the process of Photosynthesis, which acts as a driving force for both biotic and abiotic world. Photosynthesis produces 1700 million tonnes of dry matter per year by fixing 75 × 10^{12} Kg of carbon every year. Photosynthetic organisms use only 0.2 % of incident solar light on earth. Carbohydrates produced by photosynthesis are the basic raw material for respiration and also to produce many organic compounds. It maintains atmospheric oxygen and carbon dioxide level. Photosynthesis consumes atmospheric carbon dioxide which is continuously added by the respiration of organisms. Photosynthesis is the major endergonic reaction. In this chapter, we will study about the energy yielding process of photosynthesis and various types of energy utilization processes to produce carbohydrates.

13.1 Historical Events in Photosynthesis

- **Van Helmont** (1648) – Increase in organic substances comes from water alone by growing a Willow tree that gains weight but soil loses only 2 ounces of the original weight.
- **Stephen Hales** (1727) – Father of Plant Physiology, Plants obtain nourishment from air and light.
- **Joseph Priestley** (1772) – Performed experiments with candle, mice and Mint plant and concluded that vegetation purifies the air.
- **Jean-Ingen-Housz** (1779) – Confirmed Priestley's experiment that oxygen released by the plants is possible only in light.
- **Lavoisier** (1783) – Purifying gas produced by plants in sunlight is Oxygen (*Phlogiston*) and noxious gas produced by burning of candle (*de Phlogiston*) is Carbon di oxide.
- **Desaussure** (1804)- Explained the importance of water in the process of photosynthesis.
- **Dutrochet** (1837) – Explained the importance of Chlorophyll in Photosynthesis.